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The Top 5+ Do's, Don'ts & How's of Azure Virtual Desktop

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Focus

Azure Virtual Desktop & Log Analytics

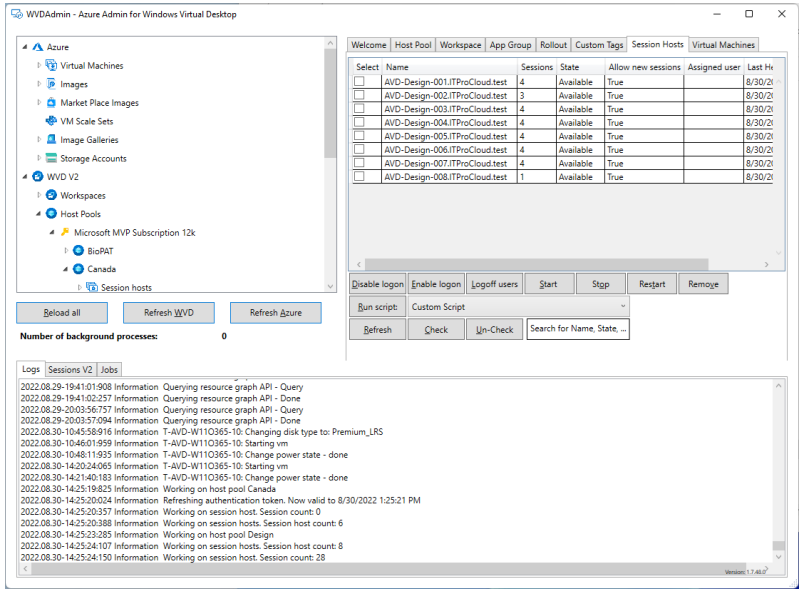
WVDAdmin & Hydra for AVD

From

Germany

My Blog

blog.ITProCloud.de



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Agenda

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- Upgrading from Windows 10 -> 11
- Azure Boost: NVMe, Ephemeral, v6
- Orphan Resources
- Imaging Strategies
- My Top 5 Pitfalls and Reminders

Key takeaways:

- Avoid failures I made before
- Keep an eye on important things
- Take away some ideas

Upgrading from Windows 10 -> 11

- Why
  - Windows 10 is EOL on October 14th, 2025

- Challenges
  - Older hosts / Golden Master on V1 disks (vTPM, Trusted Launch)
  - Windows update doesn’t show an update path
  - How to run an in-place update

  - Site-node: Strange BitLocker behaviours on some updates (also Windows 11 23H2 -> 24H2)

V1 vs. V2 VM Sizes – Trusted Launch

- To upgrade from Windows 10 to Windows 11, we need a vTPM
- Usually, V1 VM sizes have no vTPM*

- Differences

	V1	V2
Boot	PCAT / BIOS	UEFI
Disk Controller	IDE	SCSI (v6: NVME)
Advanced Security Feature	X	Yes, like vTPM, trusted launch*

- Action
  - Migrate the VM sizes (or at least the Golden Master)

- * https://learn.microsoft.com/en-us/azure/virtual-machines/trusted-launch-existing-vm-gen-1

V1 vs. V2 VM Sizes – Trusted Launch

- Challenge
  - That is not directly possible!

- Solution
  - Create a V2 VM with Windows Server 2022 and a larger OS disk (256 GB+)

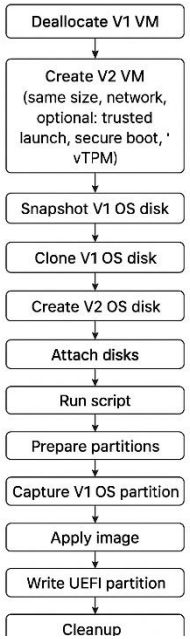
  - Create a copy of the V1 OS Disk (V1-disk)
  - Create a new V2 managed disk (V2-disk)

  - Attach both to the server VM

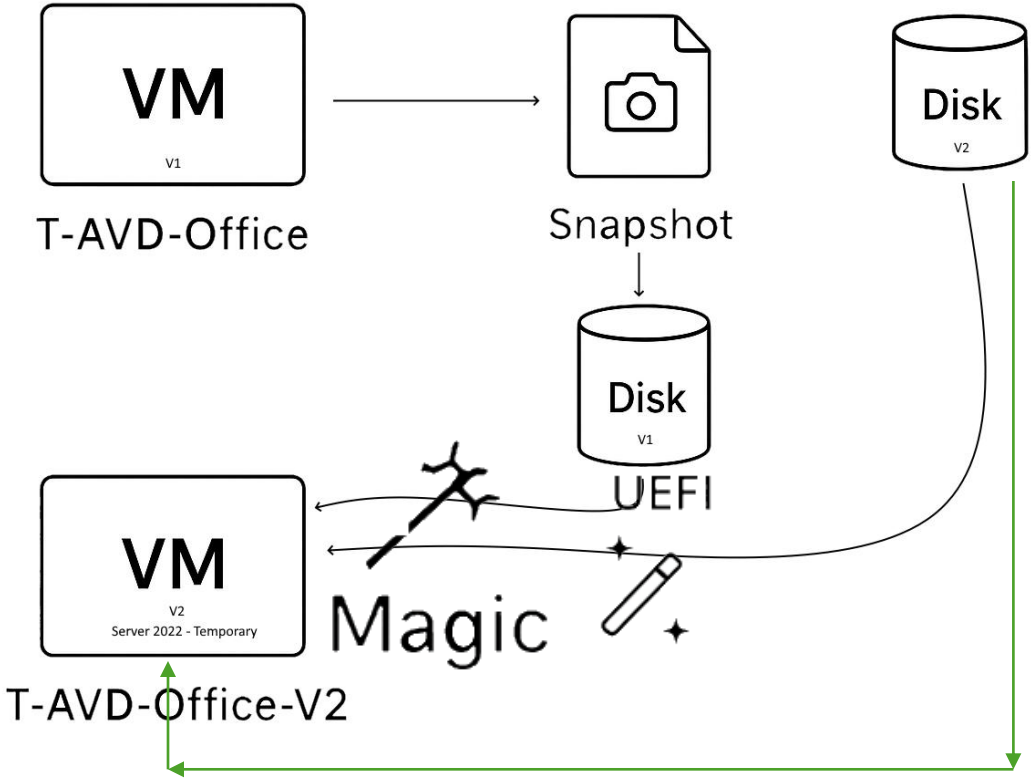
  - Capture content of the V1-disk with Dism.exe
  - Write image to the V2-disk

  - Modify partitions, write UEFI BIOS

  - We have a V2-disk with the content of the V1-disk and can create a VM



V1 vs. V2 VM Sizes – Trusted Launch

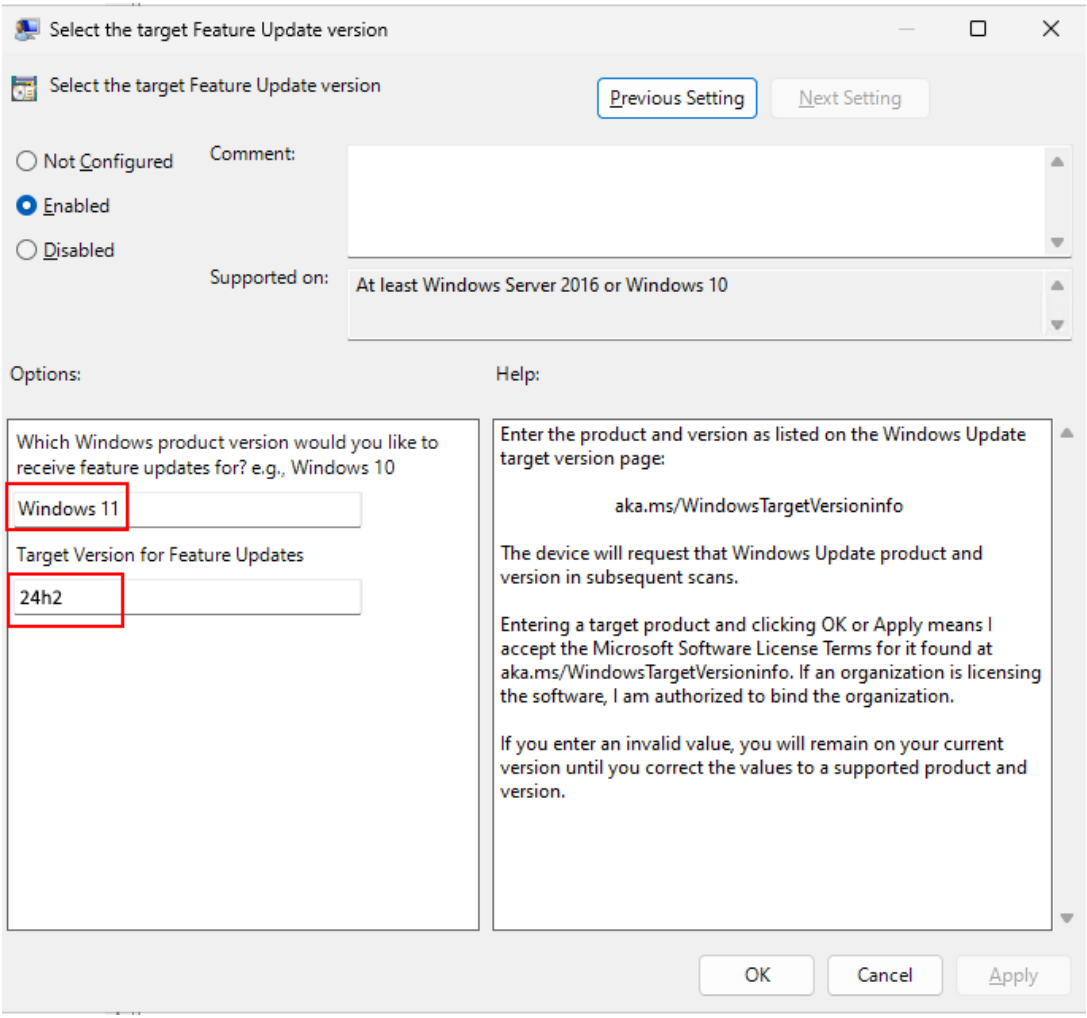


Windows – Updating

- Challenge:
  - Windows Update is not showing an update
- Two options
  - Forcing the new V2 Windows 10 client to search for a specific Windows update
  - Inplace upgrade

Windows – Updating 1

- Configuring a local policy



Windows – Updating 2

- Download multi-ISO to do an in-place upgrade
  https://www.microsoft.com/en-us/software-download/windows11

- Note: “Rumors” say that in-place upgrade is not supported in Azure and Hyper-V should be used

Windows – Ready

- We hopefully migrated our Golden Master to Windows 11
- Does this work always?
- Automate the process with PowerShell:
  - https://blog.itprocloud.de/Migrate-Azure-V1-VM-to-Azure-V2-VM-In-A-Safe-Way/
  - https://blog.itprocloud.de/Update-Windows-10-Multi-user-to-Windows-11-Multi-user/

Azure Boost: NVMe, Ephemeral on v6

The new start in town “boost VMs”

- Advantages of boost with NVMe:
  - Higher input/output operations per second (IOPS)
  - Higher throughput (MBps)
  - Faster network connection
  - Modern CPU
  - Offloading of virtualisation tasks

Azure Boost: NVMe, Ephemeral on v6

The new start in town “boost VMs”

OS disk — Premium Disk

Swap OS disk

Disk name	Storage type	Size (GiB)	Max IOPS	Max throughput (...	Encryption
AVD-Design-001_disk1_def52e63fb	Premium SSD LRS	128	500	100	SSE with PM

OS disk — Ephemeral Disk

Swap OS disk

Disk name	Storage type	Size (GiB)	Max IOPS	Max throughput (...	Encryption
AVD-Design-003_OsDisk_1_c2dc69	Standard HDD LRS	128	36000	477	SSE with PM

Azure Boost: NVMe, Ephemeral on v6

Azure Boost: NVMe, Ephemeral on v6

Challenges

- No migration path for Windows
- Images are not compatible (custom, gallery)
- Pagefile on C:\ instead of D: \, if present

Azure Boost: Migration

“There is no support way to migrate to a V6 directly.”

(as I know)

Azure Boost: Images

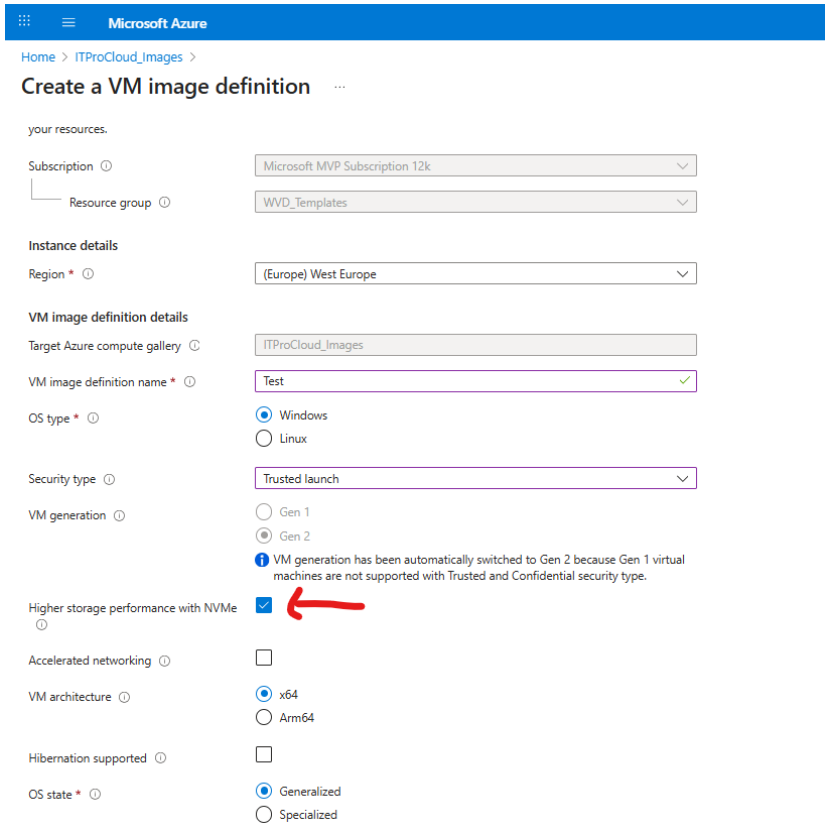
- Microsoft Marketplace images are working (scsi, nvme)
- Custom images are not working
- If we have no way to migrate a Golden Master, we have to create a new v6 Golden Master – with all the effort
- But …

Azure Boost: Images

We can use a specific Azure Compute Gallery Definition

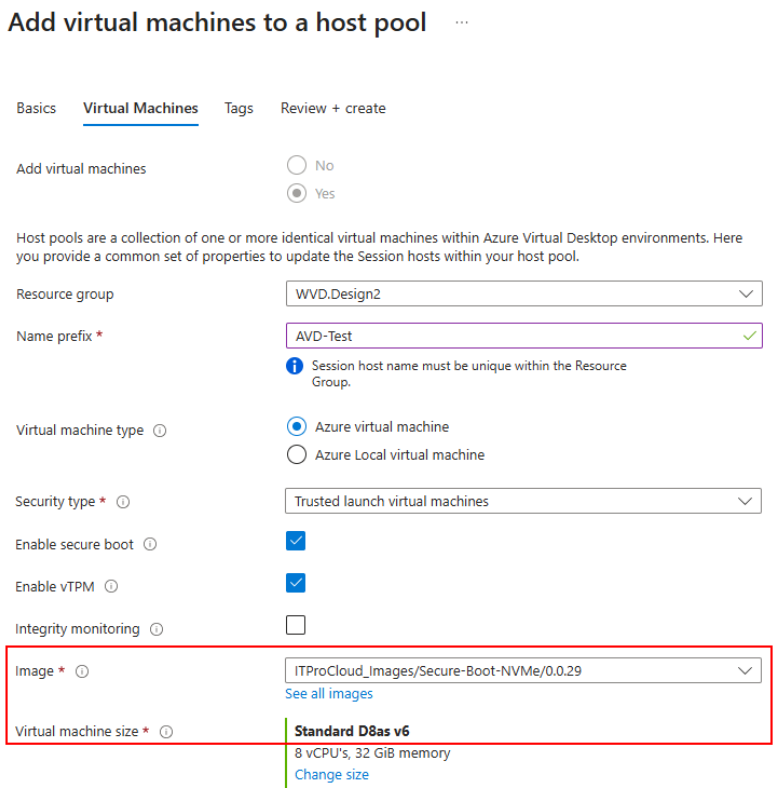
- Create a new definition
- Enable NVME
- Security type “trusted launch” (recommend)

- Capture the image from your old Golden Master into the definition (WVDAdmin, Hydra, PowerShell, …)



Azure Boost: Images

You can now deploy session hosts with that image from the Gallery Definition



Hints:

- The operating system must support NVMe

Demo Time

Azure Boost: Migration

One step back

If the OS supports NVMe, we can migrate existing VMs:

- Capture a specialized image into a prepared Gallery Definition
- Create a new v6 VM based on that image
- Done

Azure Boost: Pagefile

Do you know the “Don’t store data on D: drive” drive?

- E.g.: D8ads_v6
  - d stands for a local drive on the hypervisor
  - Perfect for the pagefile, because no network in between

And?

Hint: Check out https://cloudprice.net/

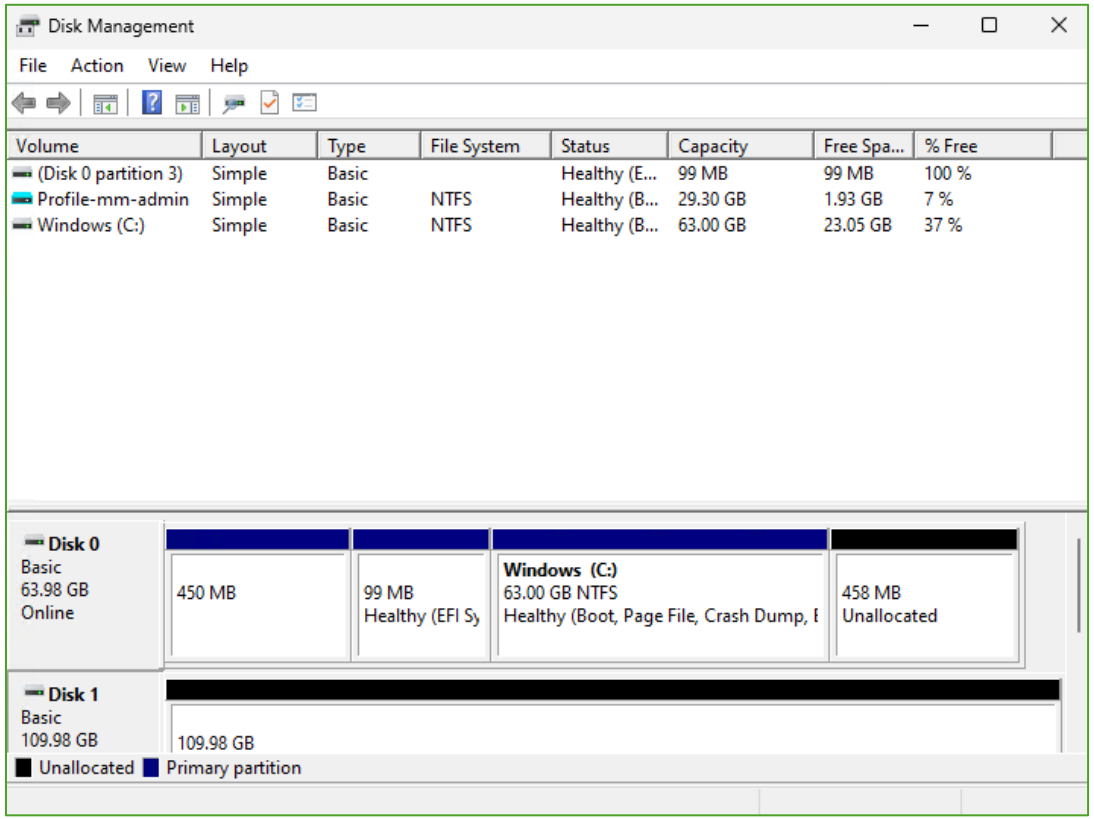
Azure Boost: Pagefile

Do you know the “Don’t store data on D: drive” drive?

The D: drive is not automatically mounted

Challenge:

- Mount D:
- Format D:
- Reconfigure the pagefile
- Reboot



Azure Boost: Pagefile

But:

C:\Windows\system32\diskpart.exe

```
DISKPART> select disk 1

Disk 1 is now the selected disk.

DISKPART> detail disk

Microsoft NVMe Direct Disk v2
Disk ID: {99F71161-C07A-48E2-A6AD-3B07C20AD2EB}
Type   : NVMe
Status : Online
Path   : 0
Target : 0
LUN ID : 0
Location Path : ACPI(_SB_)#ACPI(VMOD)#ACPI(VMBS)#VMBUS({44c4f61d-4444-4400-9d52-802e27ede19f}#{6f26848a-48d5-4f1e-8ca0-a
2c31d72aeee})#VPCI(0000)#NVME(P00T00L00)
Current Read-only State : No
Read-only  : No
Boot Disk  : No
Pagefile Disk  : No
Hibernation File Disk  : No
Crashdump Disk  : No
Clustered Disk  : No

There are no volumes.

DISKPART>
```

Azure Boost: Pagefile

- How to handle the pagefile:
- Each time the host is allocated, the id changed
- Therefore:
  - Detect if the NVMe disk exists but is not mounted/formatted
  - Mount and format the disk
  - Reconfigure pagefile
  - Do a reboot (a reboot will not change the device id)
  - Do that very early before the AVD agent starts

- The upcoming Hydra version will do that automatically if you want.

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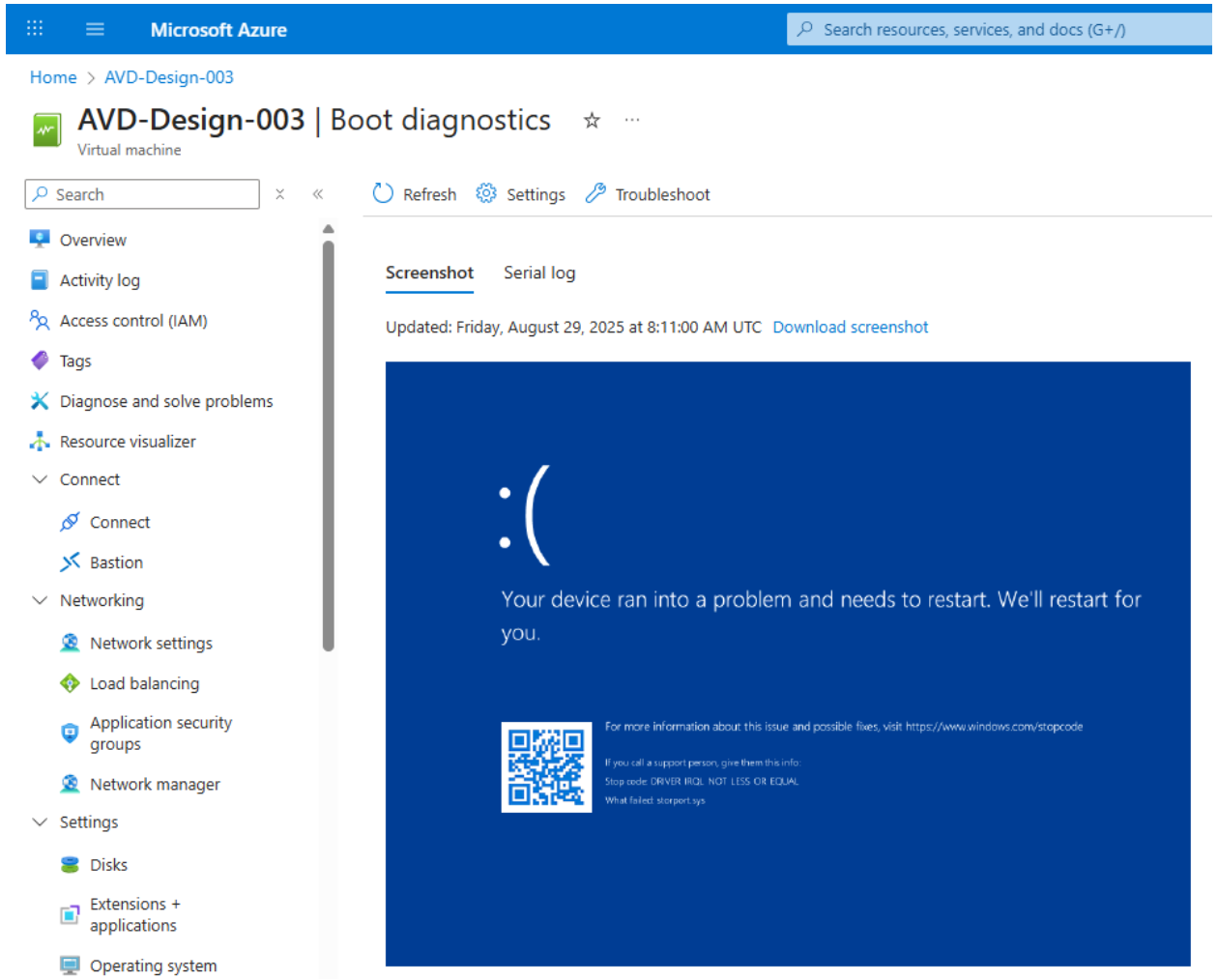
Azure Boost: Finally



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Orphan resources


```
"Name": "Design/AVD-Design-002.ITProCloud.test",
"Id": "/subscriptions/dcdce2ee-c9c0-4765-acd5-197126d21978/resourcegroups/WVD.Design2/providers/Microsoft.DesktopVirtualization/hostpools/Design/sessionhosts/AVD-Design-002
"Type": "Microsoft.DesktopVirtualization/hostpools/sessionhosts",
"Properties": {
  "LastHeartBeat": "2025-09-03T06:47:28.62Z",
  "Sessions": 0,
  "AgentVersion": "1.0.11802.2200",
  "AllowNewSession": true,
  "AssignedUser": null,
  "FriendlyName": null,
  "Status": 2,
  "StatusTimestamp": null,
  "OsVersion": "10.0.22621.5768",
  "SxSStackVersion": "rdp-sxs250508650",
  "UpdateState": 3,
  "LastUpdateTime": "2025-09-01T08:18:35.25Z",
  "UpdateErrorMessage": "",
  "SessionHostHealthCheckResultObject": [],
  "ResourceId": "/subscriptions/dcdce2ee-c9c0-4765-acd5-197126d21978/resourceGroups/WVD.Design2/providers/Microsoft.Compute/virtualMachines/AVD-Design-002",
  "VirtualMachineId": "e2330658-aeaa-46d6-8906-24cb9e14fb63"
```

Orphan resources

While a host is only linked to a VM, the following can happen:

- Orphan Host: The VM is deleted, but not the host -> Issues in autoscaling
- Orphan VM: The session host is deleted (in the Azure Portal), but the VM still exists:
  - If the VM is running, you pay for the full VM – Autoscale will not shut down the VM
  - You may run out of IP addresses
  - Storage costs
- Different and similar: If a host/VM is offline for 90+ days, the AVD Agent cannot connect to the backend the next time
  - The hosts show up as unavailable/shutdown -> See orphan host

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Orphan resources - Investigate

Azure Virtual Desktop - Deep-Insights

This workbook gives you an insight into the logs coming from the Azure Virtual Backplane. These data can be used to resolve issues while user connecting or running sessions.

Check for an update of this Workbook - Current version: v3.5

Check-out the ITProCloud Blog

- blog.itprocloud.de
- WVDAdmin
- Hydra

Hydra

mm-admin@ITProCloud.de

Message

Orphan session hosts detected (Demo Tenant): We detected some session hosts without a linked virtual machine (or offline for longer than 90 days). That is expected during a rollout or deletion. Please validate the listed Session Hosts / VMs to avoid issues with auto scaling: AVD-AAD-GPU-002, AVD-Crazy2-0.ITProCloud.test, AVD-Crazy2-1.ITProCloud.test, AVD-Crazy2-2.ITProCloud.test, AVD-Crazy2-3.ITProCloud.test, AVD-SPLIT-S01.ITProCloud.test, AVD-SPLIT-S02.ITProCloud.test, AVD-SPLIT-S03.ITProCloud.test, VDI-11-MXE-001.ITProCloud.test

Orphan session hosts detected (Azure Local Tenant): We detected some session hosts without a linked virtual machine (or offline for longer than 90 days). That is expected during a rollout or deletion. Please validate the listed Session Hosts / VMs to avoid issues with auto scaling: AVD-PROD-M-302.FRA0204-AVD-LAB.local

Orphan AVD Agents detected (Demo Tenant): We detected some running Virtual Machines that are not listed as available on Azure Virtual Desktop. That is expected during a rollout or a longer operation, like Windows Update. Please validate the listed VMs/Session Hosts to avoid additional costs and issues: AVD-LAB-001 (VM running)

led, !region

0
0
0

Warning	SPLIT-POOL	AVD-SPLIT-S02.ITProCloud.test	0	29			avd-split-s02
Warning	SPLIT-POOL	AVD-SPLIT-S01.ITProCloud.test	0	29			avd-split-s01
Warning	Demo-GPU	AVD-AAD-GPU-002	1	127	AVD-AAD-GPU-002	northeurope	avd-aad-gpu-002
Warning	Crazy-II	AVD-Crazy2-3.ITProCloud.test	0	283			avd-crazy2-3
Warning	Crazy-II	AVD-Crazy2-2.ITProCloud.test	0	283			avd-crazy2-2
Warning	Crazy-II	AVD-Crazy2-1.ITProCloud.test	0	283			avd-crazy2-1
Warning	Crazy-II	AVD-Crazy2-0.ITProCloud.test	0	283			avd-crazy2-0

Workbook/

Orphan resources

We will see it in my don’t does….

Demo Time

My top 5

Does and don’t do:

“Never delete a host in the Azure Portal without reading the small letters”

Don’t forget to delete the VM – or use fancy tools to do that. Don’t forget the disk and nic

My top 5

Does and don’t do:

“Separate FSLogix profiles and don’t mix them up (usually)”

Keep a folder for each host pool to store the profiles. Use the same name as the pool and configure NTFS permissions correctly

My top 5

Does and don’t do:

“Choose the right storage for FSLogix”

Huge differences between performance, provisioned and used capacity, pay-as-you-go, and the “Provisioned price model”

E.g., Pay-as-you-go, standard storage: Tier and transactions matters

My top 5

Does and don’t do:

“Always configure diagnostic settings on the pools”

This gives you so many insights into what happens on the backend.

https://blog.itprocloud.de/AVD-Azure-Virtual-Desktop-Error-Drill-Down-Workbook/

My top 5

Does and don’t do:

“Check for orphan resources”

Stop paying for VM of “unavailable” session hosts.

https://blog.itprocloud.de/AVD-Azure-Virtual-Desktop-Error-Drill-Down-Workbook/

My top 5

Does and don’t do:

“Don’t put a half-installed Windows Update in an Image”

Stop slow deployments.

We love Feedback

Great Session!

Okay Session!

Not so okay Session!

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Thank You